

Potential Wireless LAN Troubleshooting Causes

	Wired/Wireless	Location	Potential Issues	
1	Wireless	End User	Skills, Knowledge Perceptions, Device on/off, Understanding of Concepts & Device capabilities, Wi-Fi vs Cellular	
2		Mobile	Wi-Fi Client Device	Drivers, Radio Capabilities, Authentication Profiles, Supported PHY, QoS, Power Save, Applications, Location, MPTCP, Vendor IE Support, Chipset Behavior, Roaming Algorithms, Auto-Negotiated MCS, MDM, Protection
3			RF Media	RSSI, SNR, SNIR, Primary & Secondary Coverage, CCI/ACI, Retry Rates, Average MCS, Jitter, Latency, Consistency, Regulatory Domains, Non-Wi-Fi Interference, Spectrum Analysis
4		Per Frame Tx	Contention Process	Preamble Detect, Energy Detect, Triggers, NAV Timers, TxOP, AIFS, Random Slots, QoS, WMM, Duration ID, Ch Capacity, Non-Wi-Fi Inteference
5		Per Frame Tx	MCS Process	Per Frame Decisions - Modulation Technique, Coding Technique, Ch Width, Guard Interval, Spatial Streams, Tx Power, ACK vs No ACK, TX decides
6		Per Time	DFS Process	802.11 is NOT primary User - AP Scans for 60-Seconds, AP Enabled, Continuous Scanning, If RADAR detected, send CSA, Change to new CH, After 30-min can return, after 60-second scan
7		Per Frame Tx	Single Frame on RF	Overhead to delivery IP Payload - AIFS, CW, BPSK Preamble, RTS, SIFS, Preamble BPSK, CTS, SIFS, Preamble, Preamble VHT, Header MBR, Payload PHY rate, CRC, SIFS, Preamble, ACK
8		Per Timers	Association Process	Beacon, Probe Request, Probe Reponse, Authentication Request, Authentication Response, Association Request, Association Response, Decide on which AP by: RSSI, SNR, Auth Method, Encrypt Method, Channel Switch Announcement, Error Ratios, MCS/Data Rates Supported, Heuristics, Internal Lists, De-Authentication, Dis-Associate, 802.11 k, v, r, MBR, Proprietary Methods!
9			802.11 k, v, r	AP's try to influence the roaming decisions via 'standard' modes
10		Per Changes	Authentication Process	Open, Pre-Shared Key, 802.1X RADIUS, PSK includes Exchange of 4-Way Handshake to trigger Encryption Keys, 802.1X EAP Exchange, ending in 4-Way Handshake
11			Encrytion Process	None, TKIP, AES/CCMP, Punishment for using TKIP, Confusion with Wi-Fi Alliance naming - WPA2 PSK... is PSK-WPA2
12		From LAN	Upper Layers	DHCP, IP , DNS, VLAN, Subnet Mask, Default Gateway, Captive Portal
13			Controlled Port	AP Controls which 802.11 Frames can cross Wireless to Wired Boundary
14	Fixed	Access	Configurations, SSIDs, Minimum Basic Rates, Supported PHY Rates, Band Steering, Client Control, Radio Capabilities, Tx Rates, Client Isolation, Roaming, QOS	
	Wired	Point	PoE, Antenna Pattern, Mounting, 1GB backhaul limit, AP Locations, Physcial Layer Issues, Firmware Revisions, Custom Configurations, RRM/ARM, Proprietary	
15		Local Network	Cable Media	EIA/TIA 568A/B, Category Mismatch, Validation Tests, Grounding, other issues
16			Edge Switch	VLANs, Port Speeds, PoE, Configurations, QoS, End-to-End?, COS vs DSCP
17			Local Network	Distributed vs Centralized Forwarding, ACLs, VLANs, QoS, Tunnels, Layers, NAT
18			TCP/UDP	Following all TCP issues as well as UDP reasons for using each
19			Quality of Service	Access Port vs Trunk Port, DSCP, WMM Categories, End-to-End QoS
20			Applicatons	MTU, TCP Window, Round Trip Time, Processing Time, TCP Retransmission times
21			DHCP Server	Lease Durations, Configurations, Broadcast Storms, Latency, Performance, Address Pool Scopes, Scalability, DHCP Options, Auto Renew
22			DNS	Configuration, Scalability, Security, Accuracy, Customization, Control, Blacklists
23			802.1X/RADIUS	Configuration, Ports, Ranges, Licensing Issues, EAP types, Custom VSA, Scalability, Resouces, Certificate Issues, Fast/Secure Roaming types
24			Active Directory	Accounts, Credentials, EAP Compatibility, Custom RADIUS Attributes
25			Controller Functions	Code Versions, Bugs, Configurations, Local vs Cloud, Licensing Issues, Distributed vs Centralized Forwarding, VLAN choices
26			Firewall	Firewall Rules, Capacity, Compatibility, Rate Limiting, Bandwidth Shaping
27		WAN Router	Size of Internet Pipe, Inernet Destination Issues, Costs, Availability, Consistency	
	Internet	Internet Connection	Bandwidth Throttling, Jitter, Latency	
28		Captive Portal	Security, Client Issues, Privacy, Friction, Triggers, Certificates, DNS, Captive Portal Location, Control, Monetization, Legal, MiFi	